

# A Machine-Verified Disproof of TLMC Conjecture #2308: The Smallest Solvable Complete Group

Batch 6 solution submission

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## Abstract

We disprove a conjecture of the Last Math Competition asserting that the smallest nontrivial solvable complete group has order  $2^{11} \cdot 3 = 6144$ . The symmetric group  $S_3$  is nontrivial, solvable, and complete (trivial center, every automorphism inner), and has order  $6 \neq 6144$ . In fact  $S_3$  is the smallest nontrivial complete group, solvable or not. The formalization in Lean 4 (`Batch6.lean`) verifies all finite facts by kernel computation (`decide`) and the structural parts by Mathlib's `MulAut/IsSolvable` API; compilation is clean and the axioms are the standard three.

## 1 The conjecture

**Conjecture 1.1** (TLMC-00000002308). A complete group satisfies  $G = \text{Aut}(G)$ . The order of the smallest nontrivial solvable complete group is  $2^{11} \cdot 3$ ; its structure is a minimal example of a semidirect product of a 2-group.

## 2 The disproof

**Theorem 2.1.**  *$S_3$  is a nontrivial solvable complete group of order 6. Hence Conjecture 1.1 is **false**: the smallest nontrivial solvable complete group has order at most 6, not  $2^{11} \cdot 3 = 6144$ .*

*Proof.*  $S_3 = \text{Perm}(\{0, 1, 2\})$  has exactly six elements  $1, s_{01}, s_{02}, s_{12}, s_{01}s_{02}, s_{02}s_{01}$  (kernel-computed). *Solvable.* The map  $\text{sgn}$  sending the three transpositions to  $-1$  and the other three elements to  $+1$  is a homomorphism (kernel-verified on all 36 pairs). Its kernel is  $\{1, s_{01}s_{02}, s_{02}s_{01}\}$ , whose elements commute pairwise (kernel-verified), so  $[S_3, S_3]$  is abelian and  $[S_3, S_3]^{(1)} = 1$ : the derived series terminates in two steps. *Complete.* The center is trivial: an element commuting with  $s_{01}$  and  $s_{02}$  is 1 (kernel-verified). For innerness, note first that automorphisms preserve the involution property  $\sigma^2 = 1, \sigma \neq 1$ , and the involutions of  $S_3$  are exactly the three transpositions (kernel-verified). Any two *distinct* transpositions are simultaneously conjugate to  $(s_{01}, s_{02})$ ; the six conjugators are exhibited explicitly, case by case, with each conjugation identity kernel-verified. Given an automorphism  $\varphi$ , the pair  $(\varphi s_{01}, \varphi s_{02})$  is such a distinct pair, so some  $k$  conjugates  $s_{01}, s_{02}$  to it. The set of  $h$  with  $\varphi(h) = khk^{-1}$  contains  $s_{01}$  and  $s_{02}$  and is closed under products and inverses; since every element of  $S_3$  is a word in  $s_{01}, s_{02}$  (the six-element enumeration),  $\varphi = \text{conj}_k$  is inner. *Order.*  $|S_3| = 6 \neq 6144$ .  $\square$

*Remark.* The group of order  $2^{11} \cdot 3$  mentioned in the conjecture may exist as *some* solvable complete group, but it is far from the smallest:  $S_3$  (order 6) already qualifies, and in fact

no smaller nontrivial group is complete (groups of order  $< 6$  are abelian or have noninner automorphisms). The claimed “minimality” is what fails.

### 3 Machine verification

`Batch6.lean` (namespace `TLMCBatch6`) formalizes the above: the enumeration, involution, centralizer, and conjugation facts are proved by `decide` (Lean kernel evaluation over the six-element group); solvability via the derived series and the parity homomorphism `sgn`; completeness via `MulAut` and an equalizer argument (`S3.complete`); the final refutation is `t1m2308` together with `t1m2308_false` ( $2^{11} \cdot 3 \neq |S_3|$ ). Compilation via `lake env lean Batch6.lean` exits 0 with zero errors and warnings; `#print axioms` reports at most `[propext, Classical.choice, Quot.sound]`; no `sorry` appears.